Song Streaming

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Introduction

There are a lot of songs and artists but not all of them became famous or well-known. The dataset used contains information about the most streamed songs on the Spotify platforms. There are many things to take into consideration when making and producing a song. In this file, we are looking at what are some things or aspects that make a song stand out and how some aspects might be related to each other.

Data set obtained from: https://www.kaggle.com/datasets/amaanansari09/most-streamed-songs-all-time

Loading data and libraries

Here we are loading and importing all the libraries and data necessary to perform analyses on the dataset.

```
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
library(ggplot2)
library(ggpubr)

Features <- read.csv("~/Downloads/archive/Features.csv")
Streams <- read.csv("~/Downloads/archive/Streams.csv")
```

Data summary

Here is a summary or a quick overview of the data that helps to get an idea of what the data is about and what we will be working with. As we can see two files will be used. In Features the data that are available are id, name, duration, energy, key, loudness, mode, speechiness, acousticness, instrumentalness, liveness, valence, tempo, and danceability of a song. For Streams we have Song, Artist, Streams..Billions. and Release.Date.

```r
summary(Features)
```

```
##       id                name              duration         energy
##  Length:100         Length:100         Min.   :1.520   Min.   :0.1850
##  Class :character   Class :character   1st Qu.:3.123   1st Qu.:0.5238
##  Mode  :character   Mode  :character   Median :3.505   Median :0.6400
##                                        Mean   :3.635   Mean   :0.6255
##                                        3rd Qu.:4.030   3rd Qu.:0.7410
##                                        Max.   :7.050   Max.   :0.9120
##       key           loudness            mode       speechiness
##  Min.   : 0.00   Min.   :-12.205   Min.   :0.00   Min.   :0.02810
##  1st Qu.: 2.00   1st Qu.: -7.104   1st Qu.:0.00   1st Qu.:0.04120
##  Median : 6.00   Median : -5.971   Median :1.00   Median :0.05940
##  Mean   : 5.34   Mean   : -6.176   Mean   :0.65   Mean   :0.09645
##  3rd Qu.: 8.25   3rd Qu.: -4.941   3rd Qu.:1.00   3rd Qu.:0.10600
##  Max.   :11.00   Max.   : -2.810   Max.   :1.00   Max.   :0.43800
##   acousticness       instrumentalness       liveness         valence
##  Min.   :0.0000255   Min.   :0.000e+00   Min.   :0.0344   Min.   :0.0612
##  1st Qu.:0.0261500   1st Qu.:0.000e+00   1st Qu.:0.0943   1st Qu.:0.3407
##  Median :0.1625000   Median :0.000e+00   Median :0.1140   Median :0.4920
##  Mean   :0.2609545   Mean   :7.848e-03   Mean   :0.1657   Mean   :0.5020
##  3rd Qu.:0.4315000   3rd Qu.:8.602e-05   3rd Qu.:0.2170   3rd Qu.:0.6630
##  Max.   :0.9450000   Max.   :4.590e-01   Max.   :0.7900   Max.   :0.9690
##      tempo         danceability
##  Min.   : 74.90   Min.   :0.3400
##  1st Qu.: 98.02   1st Qu.:0.5480
##  Median :116.86   Median :0.6715
##  Mean   :121.25   Mean   :0.6496
##  3rd Qu.:142.44   3rd Qu.:0.7592
##  Max.   :186.00   Max.   :0.9210
```

```r
summary(Streams)
```

```
##      Song              Artist          Streams..Billions. Release.Date
##  Length:100         Length:100         Min.   :1.521      Length:100
##  Class :character   Class :character   1st Qu.:1.647      Class :character
##  Mode  :character   Mode  :character   Median :1.770      Mode  :character
##                                        Mean   :1.896
##                                        3rd Qu.:2.046
##                                        Max.   :3.449
```

Exploring Data

Streaming

```
Streams %>%
  top_n(10,Streams..Billions.) %>%
  select(Song,Artist,Streams..Billions.)
```

Most and least streamed songs

```
##                 Song                            Artist Streams..Billions.
## 1    Blinding Lights                        The Weeknd              3.449
## 2       Shape of You                        Ed Sheeran              3.398
## 3       Dance Monkey                       Tones And I              2.770
## 4  Someone You Loved                     Lewis Capaldi              2.680
## 5           Rockstar   Post Malone featuring 21 Savage              2.620
## 6          Sunflower          Post Malone and Swae Lee              2.575
## 7          One Dance   Drake featuring Wizkid and Kyla              2.556
## 8             Closer The Chainsmokers featuring Halsey              2.481
## 9               Stay   The Kid Laroi and Justin Bieber              2.432
## 10          Believer                    Imagine Dragons              2.410
```

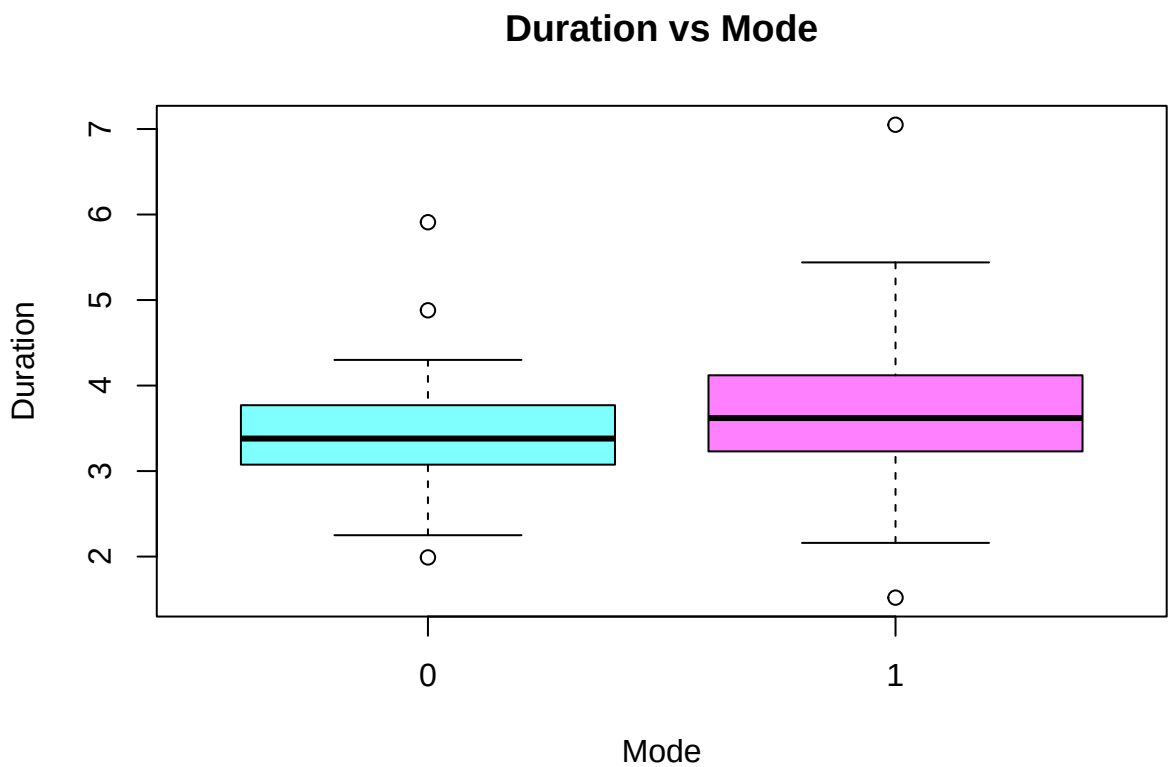
```
Streams %>%
  top_n(-1,Streams..Billions.) %>%
  select(Song,Artist,Streams..Billions.)
```

```
##                Song Artist Streams..Billions.
## 1 Don't Stop Me Now  Queen              1.521
```

In this section, we observe that all of the top 10 most streamed songs have over 2 billion streams, and the song with the least streams even though is the last has over 1 billion streams.

Boxplots

```r
boxplot(Features$duration~Features$mode,
        main = "Duration vs Mode",
        ylab = "Duration",
        xlab = "Mode",
        col = cm.colors(2))
```

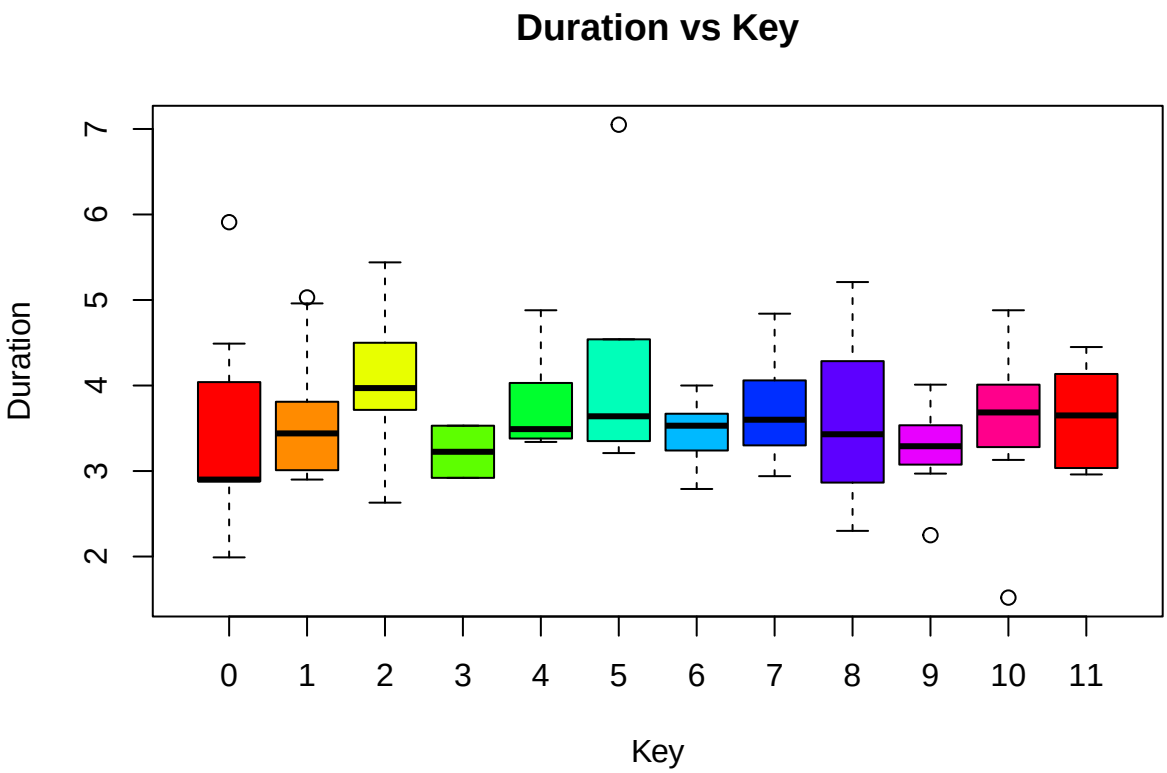



```r
aggregate(duration ~ mode, Features, mean)
```

```
##   mode duration
## 1    0 3.455714
## 2    1 3.732000
```

In this boxplot, it looks like the mean duration between mode 0 and mode 1 is about the same.

```r
boxplot(Features$duration~Features$key,
        main = "Duration vs Key",
        ylab = "Duration",
        xlab = "Key",
        col = rainbow(11))
```

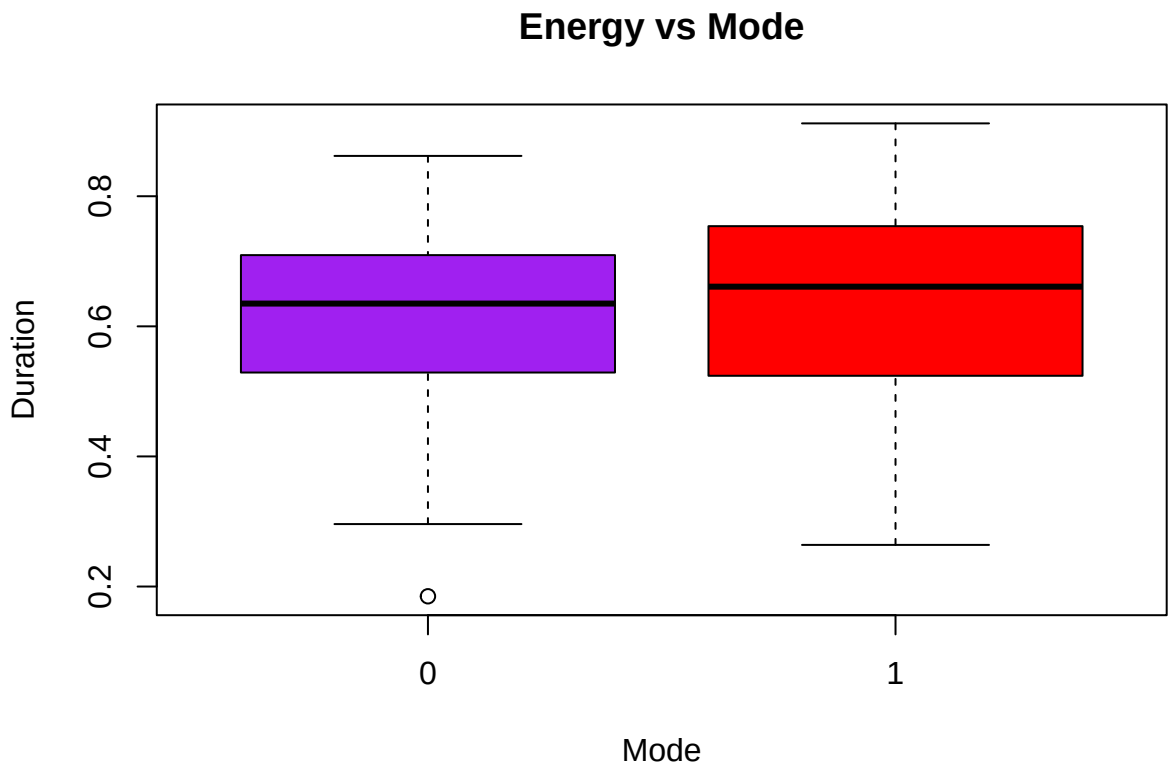



```r
aggregate(duration ~ key, Features, mean)
```

```
##    key duration
## 1    0 3.428889
## 2    1 3.586000
## 3    2 4.055000
## 4    3 3.225000
## 5    4 3.792222
## 6    5 4.358000
## 7    6 3.461818
## 8    7 3.704444
## 9    8 3.605714
## 10   9 3.252857
## 11  10 3.559000
## 12  11 3.631250
```

In this plot is shown that the mean duration between key 0 to 11 are about the same.

```r
boxplot(Features$energy~Features$mode,
        main = "Energy vs Mode",
        ylab = "Duration",
        xlab = "Mode",
        col = c("purple","red"))
```

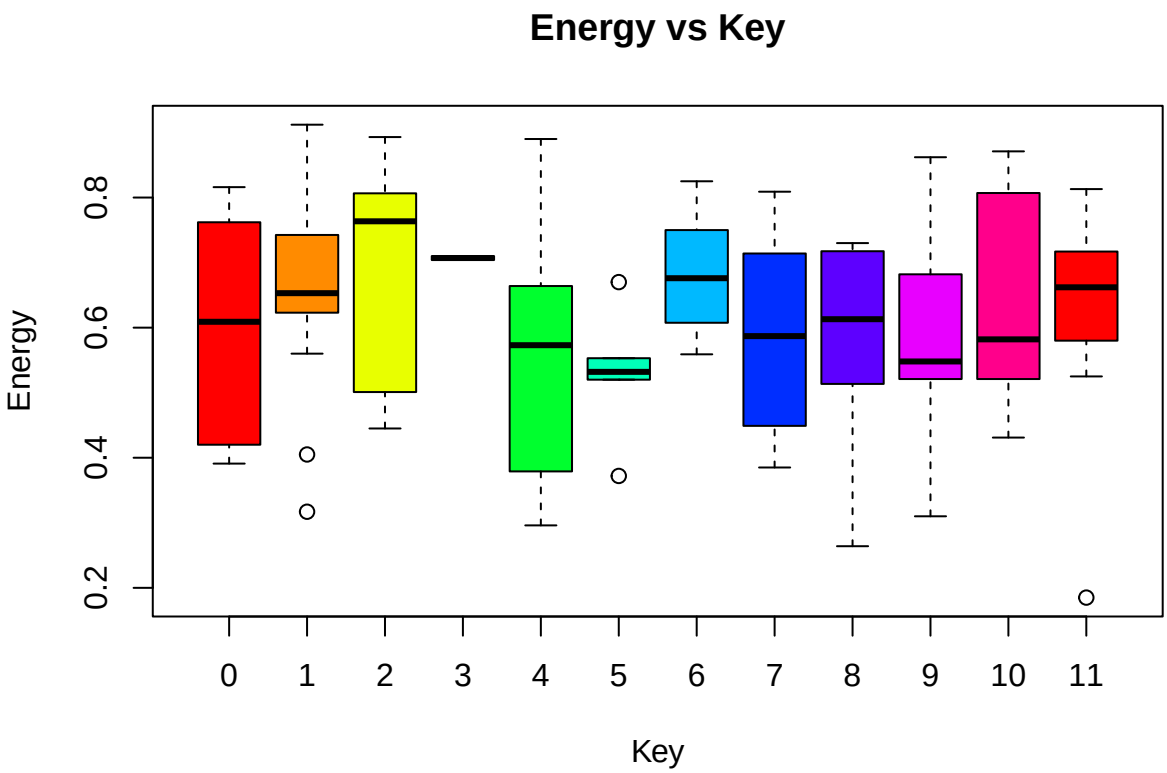



```r
aggregate(energy ~ mode, Features, mean)
```

```
##   mode    energy
## 1    0 0.6048286
## 2    1 0.6366308
```

It looks like the mean energy between mode 0 and mode 1 is about the same.

```r
boxplot(Features$energy~Features$key,
        main = "Energy vs Key",
        ylab = "Energy",
        xlab = "Key",
        col = rainbow(11))
```




```r
aggregate(energy ~ key, Features, mean)
```

```
##    key    energy
## 1    0 0.6133333
## 2    1 0.6704000
## 3    2 0.6850000
## 4    3 0.7070000
## 5    4 0.5668889
## 6    5 0.5294000
## 7    6 0.6805455
## 8    7 0.5901111
## 9    8 0.5812857
## 10   9 0.5894286
## 11  10 0.6423000
## 12  11 0.6145000
```

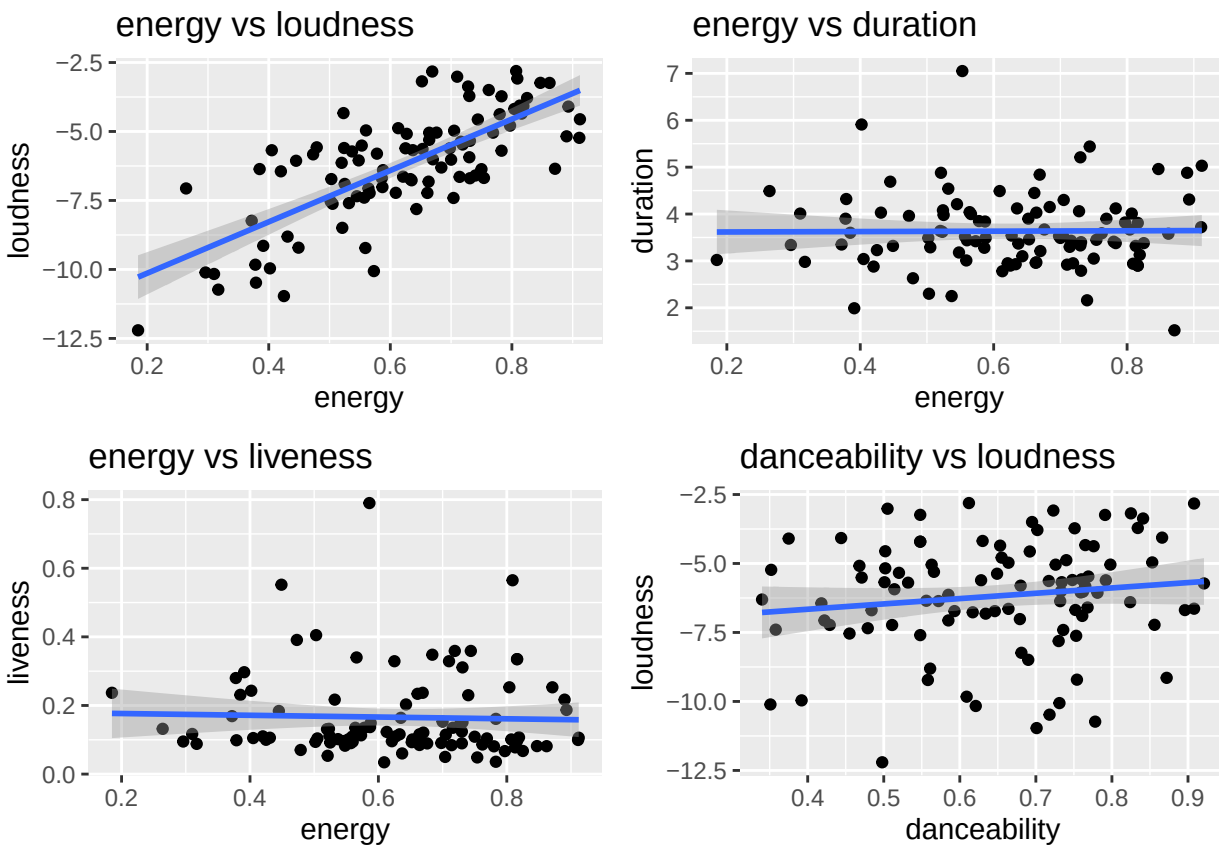
The mean energy between key 0 to 11 is overall about the same.

Linear Relationship

```
b = ggplot(Features,aes(energy, loudness)) + geom_point() +
  geom_smooth(method="lm")  + ggtitle("energy vs loudness")
c = ggplot(Features,aes(energy, duration)) + geom_point() +
  geom_smooth(method="lm")  + ggtitle("energy vs duration")
d = ggplot(Features,aes(energy, liveness)) + geom_point() +
  geom_smooth(method="lm")  + ggtitle("energy vs liveness")
e = ggplot(Features,aes(danceability, loudness)) +  geom_point() +
  geom_smooth(method="lm")  + ggtitle("danceability vs loudness")

ggarrange(b,c,d,e)
```

```
## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'
```

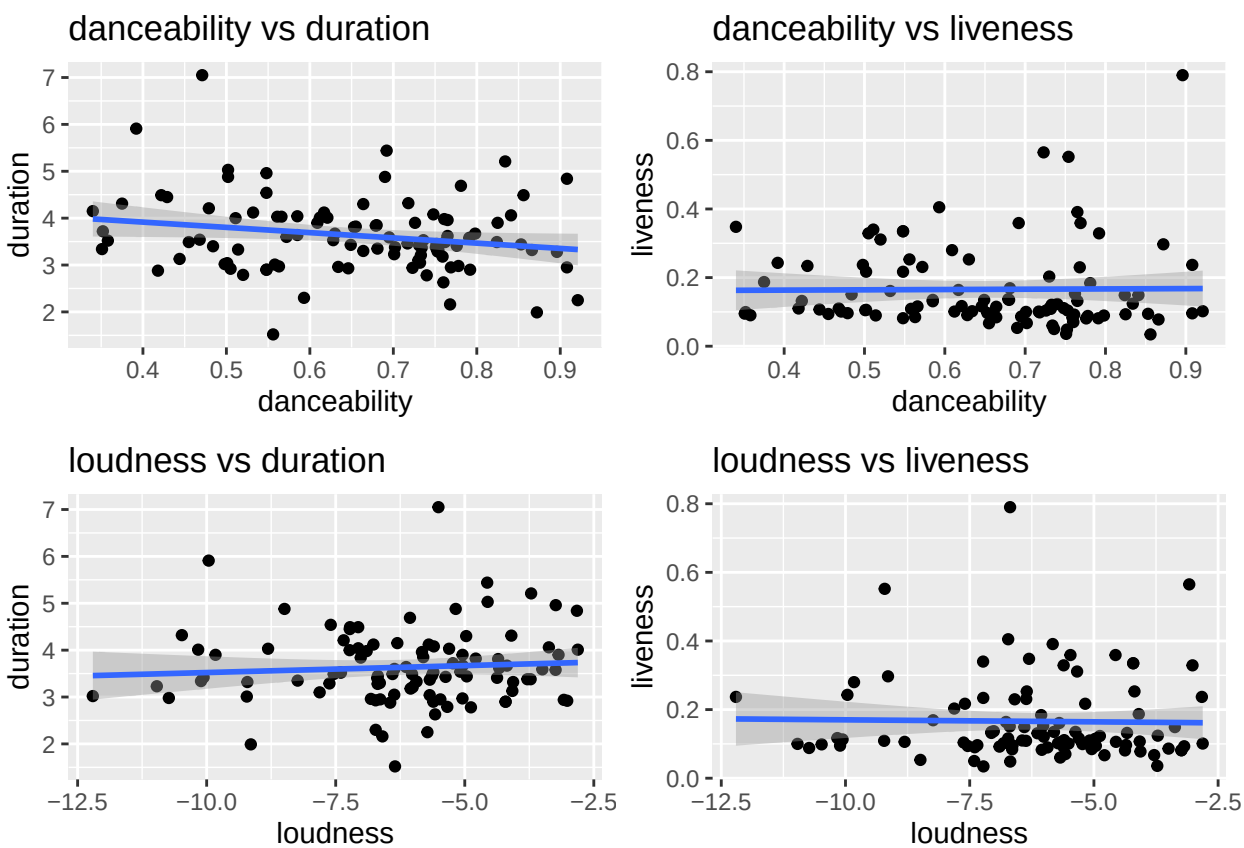


Between the different graphs shown in the portion above there seems to be a relationship between some but only the first one (loudness vs energy) seems to have a linear relation.

```r
f = ggplot(Features,aes(danceability, duration)) +  geom_point() +
  geom_smooth(method="lm") + ggtitle("danceability vs duration")
g = ggplot(Features,aes(danceability, liveness)) +   geom_point() +
  geom_smooth(method="lm") +  ggtitle("danceability vs liveness")
h = ggplot(Features,aes(loudness, duration)) +   geom_point() +
  geom_smooth(method="lm") +  ggtitle("loudness vs duration")
i =  ggplot(Features,aes(loudness, liveness)) +   geom_point() +
  geom_smooth(method="lm")  +  ggtitle("loudness vs liveness")

ggarrange(f,g,h,i)
```

```
## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'
```



On this set of graphs, the first one (duration vs danceability) is the only one that has a linear relation.

Conclusion

In this file we were able to graph and compare part of the data we have and form the data analysis and manipulation that was done, it was possible to observe that the most streamed songs had some relationships between them.